BUSFIN 8200: Asset Pricing Foundations
Problem Set 2

October 23, 2026

Question 1

1a)

Yes.

[Definition of Complete Markets will be here]

Let

$$X = \begin{pmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{pmatrix}$$

(rows: assets A, B, C; columns: states $n = 1, 2, 3$). Then there exists Ω such that $X\Omega = I$, with $\Omega = X^{-1}$, if X is invertible. Therefore, checking whether X is invertible is sufficient for Complete Markets in this economy. X has full rank. Therefore, this market is complete.

1b)

Let $P_c(n)$ denote the price of the contingent claim for state n. Then

$$P(A) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_A(n),$$
$$P(B) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_B(n),$$
$$P(C) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_C(n).$$

Or, in matrix form,

$$P = XP_c,$$

where X is the payoff matrix from 1a), $P_c = \big(P_c(1), P_c(2), P_c(3)\big)'$, and $P = \big(P(A), P(B), P(C)\big)' = (16, 8, 6)'$.

We confirmed in 1a) that the payoff matrix is invertible. So, we can calculate

$$P_c = X^{-1}P = \textit{[numerical values to be computed by the student]}.$$

1c)

As calculated in 1b), P_c = *[values from 1b)]*. Taking the inner product of $\text{Payoff}(D) = (30, 20, 10)'$ with P_c gives the price of asset D in this economy:

$$P(D) = \text{Payoff}(D)'\, P_c = \textit{[to be computed by the student]}.$$

1d)

The SDF M satisfies $\mathbb{E}[M \cdot \text{Payoff}] = \text{Price}$ for every asset in this economy:

$$\begin{aligned} P(A) &= \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_A(n), \\ P(B) &= \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_B(n), \\ P(C) &= \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_C(n). \end{aligned}$$

Or, in matrix form,

$$P = X_\pi M, \qquad X_\pi = \begin{pmatrix} 10 \cdot \frac{1}{4} & 20 \cdot \frac{1}{2} & 30 \cdot \frac{1}{4} \\ 10 \cdot \frac{1}{4} & 10 \cdot \frac{1}{2} & 10 \cdot \frac{1}{4} \\ 0 & 10 \cdot \frac{1}{2} & 10 \cdot \frac{1}{4} \end{pmatrix},$$

where $M = \big(M(1), M(2), M(3)\big)'$ and $P = \big(P(A), P(B), P(C)\big)'$.

X_π is invertible if $\det(X_\pi) \neq 0$: $\det(X_\pi)$ = *[to be computed by the student]*. Therefore, the SDF realizations are

$$M = X_\pi^{-1} P = \textit{[numerical values to be computed by the student]}.$$

This economy satisfies No-Arbitrage.

[Definition of No-Arbitrage will be here]

As in 1b), every contingent claim satisfies the No-Arbitrage condition above.

As we can see, the realized SDF is the same in every state of the economy. It may imply that each state is equally valuable *[?]*.

1e)

Using the SDF from 1d), with $\text{Payoff}_D(1) = 30$, $\text{Payoff}_D(2) = 20$, and $\text{Payoff}_D(3) = 10$,

$$P(D) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_D(n) = \textit{[to be computed by the student; compare with 1c)]}.$$

Question 2

2a)

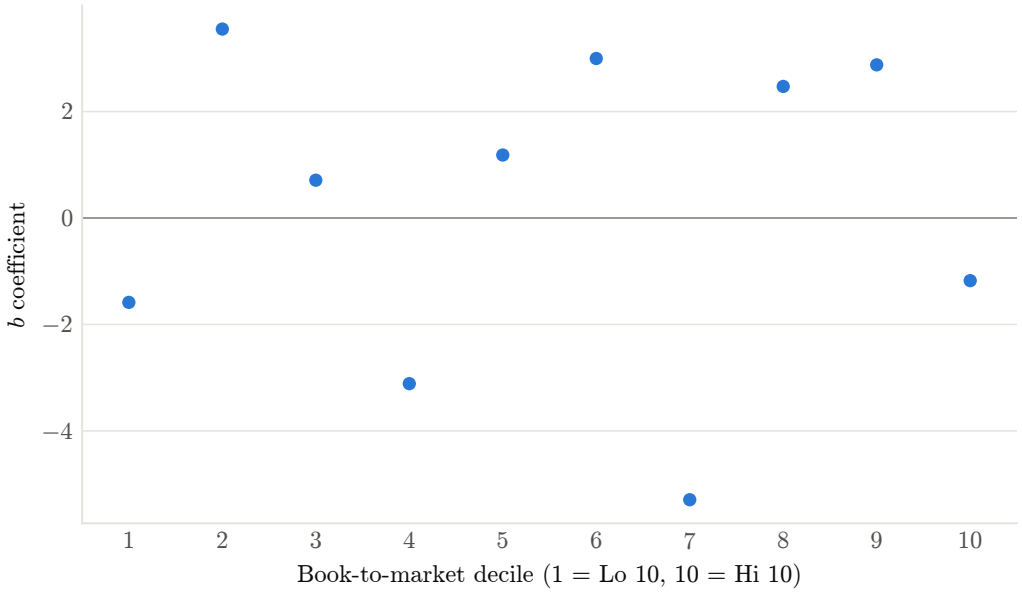


Figure 1: Coefficients b of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 10 value-weighted book-to-market decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2b)

The sample (Pearson) correlation between the monthly SDF series $M^*_{\mathrm{BM},t}$ estimated in 2a) and $M^*_{\mathrm{MOM},t}$ estimated in 2b), over July 1927 to August 2026 ($T = 1190$ months), is

$$\widehat{\mathrm{Corr}}\left(M^*_{\mathrm{BM},t},\, M^*_{\mathrm{MOM},t}\right) = 0.355.$$

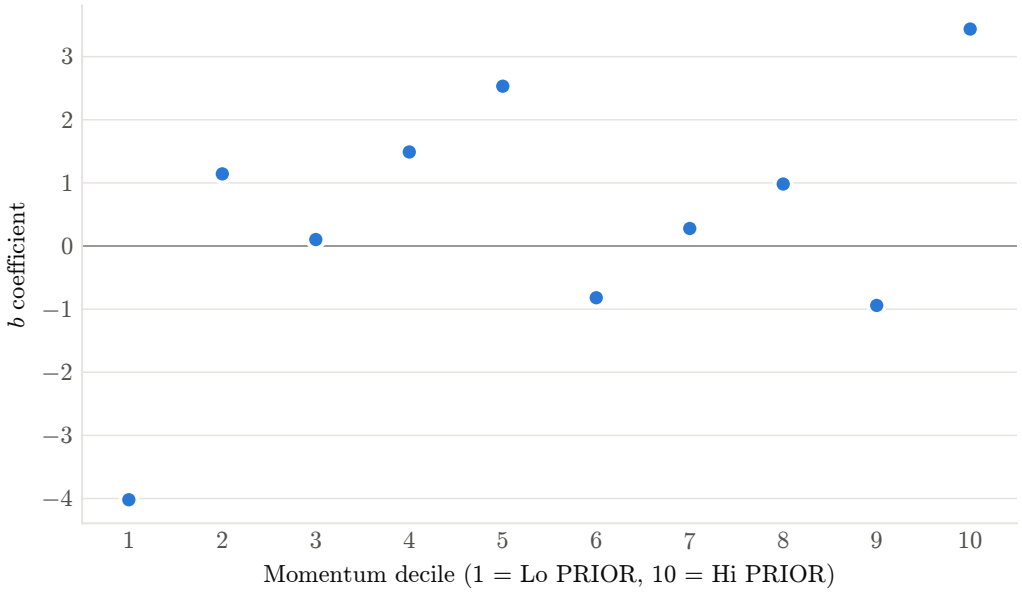


Figure 2: Coefficients b of the SDF $M_t^* = a - b' x R_t$ that prices the monthly excess returns of the 10 value-weighted momentum (prior 12–2 month return) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2c)

Table 1: Monthly Sharpe ratios $\widehat{SR}(xR^*) = \widehat{\mathbb{E}}[xR_t^*]/\widehat{\sigma}(xR_t^*)$ of the portfolios $xR_t^* = b' x R_t$ implied by the SDFs of 2a)–2c), July 1927 to August 2026 ($T = 1190$ months). Not annualized.

SDF	Test assets	$\widehat{SR}(xR^*)$
2a)	10 book-to-market deciles	0.192
2b)	10 momentum deciles	0.258
2c)	20 book-to-market and momentum deciles	0.305

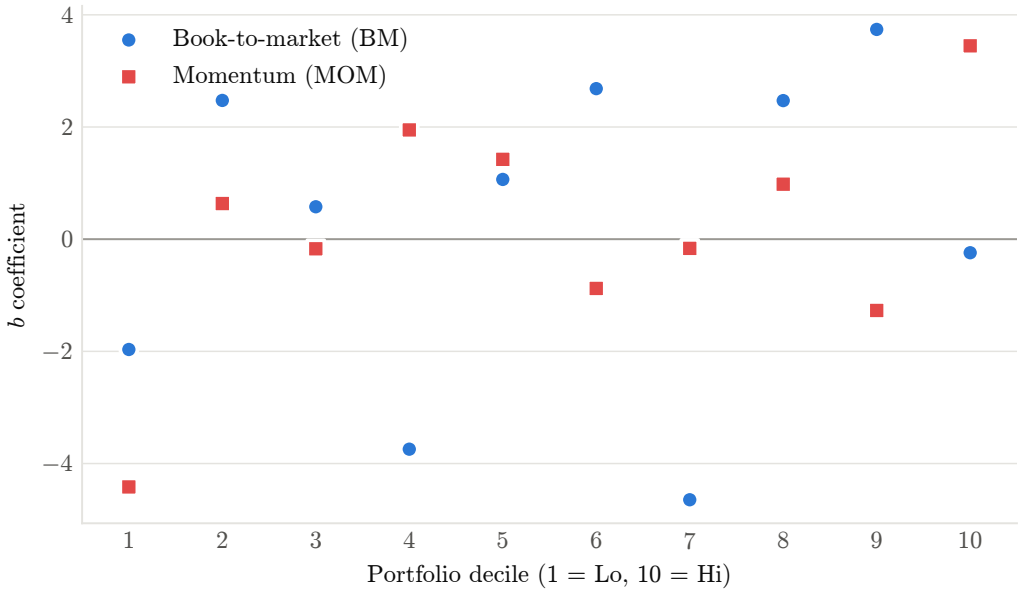


Figure 3: Coefficients b of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 20 value-weighted portfolios in the merge of the book-to-market (blue circles) and momentum (red squares) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2d)

[Explanation will be here]

	SDF M_t^* estimated in		
Portfolio	2a) BM	2b) MOM	2c) BM + MOM
Panel A: book-to-market deciles			
1 (Lo 10)	0.00	−0.15	0.00
2	0.00	−0.01	0.00
3	0.00	−0.02	0.00
4	0.00	−0.08	0.00
5	0.00	0.05	0.00
6	0.00	0.13	0.00
7	0.00	0.03	0.00
8	0.00	0.24	0.00
9	0.00	0.34	0.00
10 (Hi 10)	0.00	0.32	0.00
Panel B: momentum deciles			
1 (Lo PRIOR)	−1.02	0.00	0.00
2	−0.51	0.00	0.00
3	−0.33	0.00	0.00
4	−0.15	0.00	0.00
5	−0.09	0.00	0.00
6	−0.03	0.00	0.00
7	0.05	0.00	0.00
8	0.19	0.00	0.00
9	0.22	0.00	0.00
10 (Hi PRIOR)	0.54	0.00	0.00

Table 2: Pricing errors $\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p]$ of the 20 book-to-market and momentum decile portfolios under the SDFs M_t^* estimated in 2a) (10 book-to-market deciles), 2b) (10 momentum deciles) and 2c) (all 20 portfolios), in percent per month. Monthly excess returns, July 1927 to August 2026 ($T = 1190$ months); covariances divide by T.

2e)

[Explanation will be here]

Portfolio	$\hat{\alpha}_{OS,p}$
Panel A: book-to-market deciles	
1 (Lo 10)	1.10
2	0.79
3	0.90
4	0.70
5	0.80
6	0.72
7	0.70
8	0.59
9	0.83
10 (Hi 10)	0.97
Panel B: momentum deciles	
1 (Lo PRIOR)	2.71
2	1.65
3	1.39
4	1.21
5	0.92
6	0.83
7	0.76
8	0.64
9	0.43
10 (Hi PRIOR)	0.55

Table 3: Out-of-sample pricing errors $\hat{\alpha}_{OS,p} = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*_{OS}, xR_p]$ of the 20 book-to-market and momentum decile portfolios, in percent per month. $\hat{a}$ and $\hat{b}$ are estimated as in 2c) on July 1927 to December 1989 ($T = 750$ months); $M^*_{OS,t} = \hat{a} - \hat{b}' xR_t$ and the moments are computed on January 1990 to August 2026 ($T = 440$ months); covariances divide by T.

Question 3

Question 4